1) a)

$y = \arcsin x$

$\Rightarrow \sin y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\sin y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \cos y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos y}$

$= \frac{1}{\sqrt{1-\sin^2 y}}$

$= \frac{1}{\sqrt{1-x^2}}$

1) b)

$y = \arccos x$

$\Rightarrow \cos y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\cos y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\sin y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sin y}$

$= -\frac{1}{\sqrt{1-\cos^2 y}}$

$= -\frac{1}{\sqrt{1-x^2}}$

1) c)

$y = \arctan x$

$\Rightarrow \tan y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\tan y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec^2 y}$

$= \frac{1}{1+\tan^2 y}$

$= \frac{1}{1+x^2}$

2) a)

$y = \arcsin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\frac{1}{3}} = \frac{1}{\sqrt{1-\left(\frac{1}{3}\right)^2}} = \frac{1}{\sqrt{\frac{8}{9}}} = \frac{3}{2\sqrt{2}} = \frac{3\sqrt{2}}{4}$

2) b)

$y = \arccos x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=-\frac{1}{4}} = -\frac{1}{\sqrt{1-\left(-\frac{1}{4}\right)^2}} = -\frac{1}{\sqrt{\frac{15}{16}}} = -\frac{4}{\sqrt{15}} = -\frac{4\sqrt{15}}{15}$

2) c)

$y = \arctan x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{1+x^2}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\frac{2}{5}} = \frac{1}{1+\left(\frac{2}{5}\right)^2} = \frac{1}{1+\frac{4}{25}} = \frac{1}{\frac{29}{25}} = \frac{25}{29}$

3) a)

$y = \arcsin 2x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times \frac{1}{\sqrt{1-(2x)^2}}$

$= \frac{2}{\sqrt{1-4x^2}}$

3) b)

$y = \arccos 3x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 3 \times -\frac{1}{\sqrt{1-(3x)^2}}$

$= -\frac{3}{\sqrt{1-9x^2}}$

3) c)

$y = \arctan 8x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 8 \times \frac{1}{1+(8x)^2}$

$= \frac{8}{1+64x^2}$

3) d)

$y = \arcsin \frac{1}{3}x$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3} \times \frac{1}{\sqrt{1-\left(\frac{1}{3}x\right)^2}}$

$= \frac{1}{3\sqrt{1-\frac{1}{9}x^2}}$

$= \frac{1}{3\sqrt{\frac{1}{9}(9-x^2)}}$

$= \frac{1}{\sqrt{9-x^2}}$

3) e)

$y = \arccos \frac{x}{5}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{5} \times -\frac{1}{\sqrt{1-\left(\frac{1}{5}x\right)^2}}$

$= -\frac{1}{5\sqrt{1-\frac{1}{25}x^2}}$

$= -\frac{1}{5\sqrt{\frac{1}{25}(25-x^2)}}$

$= -\frac{1}{\sqrt{25-x^2}}$

3) f)

$$y = \arctan\left(-\frac{2}{3}x\right)$$

$$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2}{3} \times \frac{1}{1+\left(-\frac{2}{3}x\right)^2}$$

$$= -\frac{2}{3\left(1+\frac{4}{9}x^2\right)}$$

$$= -\frac{2}{3+\frac{4}{3}x^2}$$

$$= -\frac{6}{9+4x^2}$$

4) a)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\arcsin\frac{1}{2}x\right) = \frac{1}{2} \times \frac{1}{\sqrt{1-\left(\frac{1}{2}x\right)^2}}$$

$$= \frac{1}{2\sqrt{1-\frac{1}{4}x^2}}$$

$$= \frac{1}{2\sqrt{\frac{1}{4}(4-x^2)}}$$

$$= \frac{1}{2\times\frac{1}{2}\sqrt{4-x^2}}$$

$$= \frac{1}{2\sqrt{4-x^2}}$$

4) b)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\arccos\frac{3}{4}x\right) = \frac{3}{4} \times -\frac{1}{\sqrt{1-\left(\frac{3}{4}x\right)^2}}$$

$$= -\frac{3}{4\sqrt{1-\frac{9}{16}x^2}}$$

$$= -\frac{3}{4\sqrt{\frac{1}{16}(16-9x^2)}}$$

$$= -\frac{3}{4\times\frac{1}{4}\sqrt{16-9x^2}}$$

$$= -\frac{3}{\sqrt{16-9x^2}}$$

4) c)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\arctan\frac{5}{3}x\right) = \frac{5}{3} \times \frac{1}{1+\left(\frac{5}{3}x\right)^2}$

$= \frac{5}{3\left(1+\frac{25}{9}x^2\right)}$

$= \frac{5}{3+\frac{25}{3}x^2}$

$= \frac{15}{9+25x^2}$

5)

$y = 8\arcsin\frac{1}{4}x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 8 \times \frac{1}{4} \times \frac{1}{\sqrt{1-\left(\frac{1}{4}x\right)^2}}$

$= \frac{2}{\sqrt{1-\frac{1}{16}x^2}}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=2} = \frac{2}{\sqrt{1-\frac{1}{16}(2)^2}}$

$= \frac{2}{\sqrt{\frac{3}{4}}} = 2 \times \frac{2}{\sqrt{3}} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$

6)

$y = 3\arccos x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{3}{\sqrt{1-x^2}}$

$y|_{x=\frac{1}{2}} = 3\arccos\frac{1}{2} = \frac{\pi}{3}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\frac{1}{2}} = -\frac{3}{\sqrt{1-\left(\frac{1}{2}\right)^2}} = -\frac{3}{\sqrt{\frac{3}{4}}} = -3 \times \frac{2}{\sqrt{3}} = -\frac{6}{\sqrt{3}} = -2\sqrt{3}$

So the equation of the tangent will be:

$y - \frac{\pi}{3} = -2\sqrt{3}\left(x - \frac{1}{2}\right)$

$\Rightarrow y = -2\sqrt{3}x + \sqrt{3} + \frac{\pi}{3}$

7)

$y = 2\arccos\frac{x}{3} - \arcsin\frac{x}{3}$

$\frac{dy}{dx} = -2 \times \frac{1}{3} \times \frac{1}{\sqrt{1-\left(\frac{x}{3}\right)^2}} - \frac{1}{3} \times \frac{1}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$

$= -\frac{2}{3\sqrt{1-\frac{1}{9}x^2}} - \frac{1}{3\sqrt{1-\frac{1}{9}x^2}}$

$= -\frac{1}{\sqrt{1-\frac{1}{9}x^2}}$

$y|_{x=\frac{3}{\sqrt{2}}} = 2\arccos\left(\frac{\frac{3}{\sqrt{2}}}{3}\right) - \arcsin\left(\frac{\frac{3}{\sqrt{2}}}{3}\right) = 2\arccos\frac{1}{\sqrt{2}} - \arcsin\frac{1}{\sqrt{2}} = 2 \times \frac{\pi}{4} - \frac{\pi}{4} = \frac{\pi}{4}$

$\left.\frac{dy}{dx}\right|_{x=\frac{3}{\sqrt{2}}} = -\frac{1}{\sqrt{1-\frac{1}{9}\left(\frac{3}{\sqrt{2}}\right)^2}} = -\frac{1}{\sqrt{\frac{1}{2}}} = -\sqrt{\frac{2}{1}} = -\sqrt{2}$

So the equation of the tangent will be:

$y - \frac{\pi}{4} = -\sqrt{2}\left(x - \frac{3}{\sqrt{2}}\right)$

$\Rightarrow y = -\sqrt{2}x + 3 + \frac{\pi}{4}$

8)

$y = 4\arctan\left(\frac{2x}{5}\right)$

$\frac{dy}{dx} = 4 \times \frac{2}{5} \times \frac{1}{1+\left(\frac{2x}{5}\right)^2}$

$= \frac{8}{5\left(1+\frac{4}{25}x^2\right)}$

$= \frac{8}{5+\frac{4}{5}x^2}$

$= \frac{40}{25+4x^2}$

$y|_{x=\frac{5}{2}} = 4\arctan\left(\frac{2}{5} \times \frac{5}{2}\right) = 4\arctan 1 = 4 \times \frac{\pi}{4} = \pi$

$\left.\frac{dy}{dx}\right|_{x=\frac{5}{2}} = \frac{40}{25+4\left(\frac{5}{2}\right)^2} = \frac{40}{25+25} = \frac{4}{5}$

So the equation of the tangent will be:

$y - \pi = \frac{4}{5}\left(x - \frac{5}{2}\right)$

$\Rightarrow y = \frac{4}{5}x + \pi - 2$

9)

$y = 4 - 3\arctan\frac{x}{4} - 2\arctan\frac{x}{3}$

$\frac{dy}{dx} = -3 \times \frac{1}{4} \times \frac{1}{1+\left(\frac{x}{4}\right)^2} - 2 \times \frac{1}{3} \times \frac{1}{1+\left(\frac{x}{3}\right)^2}$

$= -\frac{3}{4\left(1+\frac{1}{16}x^2\right)} - \frac{2}{3\left(1+\frac{1}{9}x^2\right)}$

$= -\frac{3}{4+\frac{1}{4}x^2} - \frac{2}{3+\frac{1}{3}x^2}$

$= -\frac{12}{16+x^2} - \frac{6}{9+x^2}$

$y|_{x=0} = 4 - 3\arctan 0 - 2\arctan 0 = 4$

$\left.\frac{dy}{dx}\right|_{x=0} = -\frac{12}{16+0} - \frac{6}{9+0} = -\frac{17}{12}$

So the equation of the normal will be:

$y - 4 = \frac{12}{17}(x - 0)$

$\Rightarrow 17y - 68 = 12x$

$\Rightarrow -12x + 17y = 68$

10)

$f(x) = \arctan x \Rightarrow f'(x) = \frac{1}{1+x^2}$

$g(x) = \arctan 3x \Rightarrow g'(x) = 3 \times \frac{1}{1+(3x)^2} = \frac{3}{1+9x^2}$

$f'(x) = g'(x)$

$\Rightarrow \frac{1}{1+x^2} = \frac{3}{1+9x^2}$

$\Rightarrow 1 + 9x^2 = 3 + 3x^2$

$\Rightarrow 6x^2 = 2$

$\Rightarrow x^2 = \frac{1}{3}$

$\Rightarrow x = \pm\frac{1}{\sqrt{3}} = \pm\frac{\sqrt{3}}{3}$

11) a)

$y = \arcsin\left(\frac{2}{3}x\right)$

$\frac{dy}{dx} = \frac{2}{3} \times \frac{1}{\sqrt{1-\left(\frac{2}{3}x\right)^2}}$

$= \frac{2}{3\sqrt{1-\frac{4}{9}x^2}}$

$= \frac{2}{3\sqrt{\frac{1}{9}(9-4x^2)}}$

$= \frac{2}{\sqrt{9-4x^2}}$

For $-\frac{3}{2} \leq x \leq \frac{3}{2}$ we have $0 \leq 9 - 4x^2 \leq 9$

So $0 \leq \sqrt{9-4x^2} \leq 3$

Which means that $\frac{dy}{dx} = \frac{2}{\sqrt{9-4x^2}} \geq \frac{2}{3}$ for all $-\frac{3}{2} \leq x \leq \frac{3}{2}$

So the gradient is always at least $\frac{2}{3}$ and is never equal to $\frac{1}{2}$

11) b)

$y = \arcsin\left(\frac{2}{3}x\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{\sqrt{9-4x^2}}$

$y|_{x=-\frac{3\sqrt{2}}{4}} = \arcsin\left(\frac{2}{3} \times \left(-\frac{3\sqrt{2}}{4}\right)\right) = \arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}$

So P has coordinates $\left(-\frac{3\sqrt{2}}{4}, -\frac{\pi}{4}\right)$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=-\frac{3\sqrt{2}}{4}} = \frac{2}{\sqrt{9-4\left(-\frac{3\sqrt{2}}{4}\right)^2}} = \frac{2}{\sqrt{\frac{9}{2}}} = \frac{2\sqrt{2}}{3}$

So the equation of the normal will be:

$y + \frac{\pi}{4} = -\frac{3}{2\sqrt{2}}\left(x + \frac{3\sqrt{2}}{4}\right)$

The normal crosses the y-axis when $x = 0$:

$y + \frac{\pi}{4} = -\frac{3}{2\sqrt{2}}\left(\frac{3\sqrt{2}}{4}\right) \Rightarrow y = -\frac{9}{8} - \frac{\pi}{4}$

So Q has coordinates $\left(0, -\frac{9}{8} - \frac{\pi}{4}\right)$

So the area of the triangle is

$= \frac{1}{2} \times \left|-\frac{9}{8} - \frac{\pi}{4}\right| \times \left|-\frac{3\sqrt{2}}{4}\right|$

$= \frac{3\sqrt{2}}{8}\left(\frac{9}{8} + \frac{\pi}{4}\right)$

$= \frac{3\sqrt{2}}{64}(9 + 2\pi)$ units^2

12)

$y = \arcsin x - \arccos x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} - \left(-\frac{1}{\sqrt{1-x^2}}\right)$

$= \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}$

$= \frac{2}{\sqrt{1-x^2}}$

$\sqrt{1-x^2} > 0$ for $-1 < x < 1$, so $\frac{2}{\sqrt{1-x^2}} > 0$

Hence the curve is increasing for $-1 < x < 1$

13) a)

$y = 3\arctan 6x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \times 6 \times \frac{1}{1+(6x)^2}$

$= \frac{18}{1+36x^2}$

Set $\frac{\mathrm{d}y}{\mathrm{d}x} = 24$

$\frac{18}{1+36x^2} = 24$

$\Rightarrow 18 = 24 + 864x^2$

$\Rightarrow x^2 = -\frac{1}{44}$ which has no real solutions

So there are no points on the curve where the gradient is equal to 24

13) b)

$1 + 36x^2 \geq 1$ for all real values of x

$1 + 36x^2 = 1$ when $x = 0$

So the maximum value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ occurs when $x = 0$

$\therefore$ Maximum gradient $= 18$

14)

$y = \arctan 2x - x^2$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times \frac{1}{1+(2x)^2} - 2x$

$= \frac{2}{1+4x^2} - 2x$

$= \frac{2-2x(1+4x^2)}{1+4x^2}$

$= \frac{2-2x-8x^3}{1+4x^2}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \frac{2-2x-8x^3}{1+4x^2} = 0$

$\Rightarrow 2 - 2x - 8x^3 = 0$

$\Rightarrow 4x^3 + x - 1 = 0$

$\Rightarrow x = \frac{1}{2}$

$y|_{x=\frac{1}{2}} = \arctan\left(2 \times \frac{1}{2}\right) - \left(\frac{1}{2}\right)^2$

$= \arctan 1 - \frac{1}{4}$

$= \frac{\pi}{4} - \frac{1}{4}$

$= \frac{1}{4}(\pi - 1)$

So there is a single stationary point at $\left(\frac{1}{2}, \frac{1}{4}(\pi - 1)\right)$

15) a)

$\mathrm{f}(x) = \arctan 3x$

$\Rightarrow \mathrm{f}'(x) = 3 \times \frac{1}{1+(3x)^2} = \frac{3}{1+9x^2}$

15) b)

$\mathrm{f}'(x) = 3(1 + 9x^2)^{-1}$

$\approx 3\left(1 + (-1)(9x^2) + \frac{(-1)(-2)}{2}(9x^2)^2 + \cdots\right)$

$= 3 - 27x^2 + 243x^4 + \cdots$

15) c)

$\arctan 3x \approx \int (3 - 27x^2 + 243x^4 + \cdots)\,\mathrm{d}x$

$= 3x - 9x^3 + \frac{243}{5}x^5 + \cdots$

16)

$y = p\arcsin x - qx$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{p}{\sqrt{1-x^2}} - q$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\frac{p}{\sqrt{1-x^2}} - q = 0$

$\Rightarrow \frac{p}{\sqrt{1-x^2}} = q$

$\Rightarrow \sqrt{1-x^2} = \frac{p}{q}$

$\Rightarrow 1 - x^2 = \frac{p^2}{q^2}$

$\Rightarrow x^2 = 1 - \frac{p^2}{q^2}$

$\Rightarrow x = \pm\sqrt{1 - \frac{p^2}{q^2}}$

If $0 < p < q$ (as p and q are both positive constants) then $p^2 < q^2$, and so $\frac{p^2}{q^2} < 1$

$\therefore 1 - \frac{p^2}{q^2} > 0$ and so $y = \mathrm{f}(x)$ has precisely two stationary points when $p < q$

17)

$y = 5x + \sqrt{21}\arccos x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5 - \frac{\sqrt{21}}{\sqrt{1-x^2}}$

The curve is increasing when $\frac{\mathrm{d}y}{\mathrm{d}x} > 0$

$5 - \frac{\sqrt{21}}{\sqrt{1-x^2}} > 0$

$\Rightarrow 5 > \frac{\sqrt{21}}{\sqrt{1-x^2}}$

$\Rightarrow \sqrt{1-x^2} > \frac{\sqrt{21}}{5}$

$\Rightarrow 1 - x^2 > \frac{21}{25}$

$\Rightarrow 0 > x^2 - \frac{4}{25}$

$\Rightarrow \left(x - \frac{2}{5}\right)\left(x + \frac{2}{5}\right) < 0$

$\Rightarrow -\frac{2}{5} < x < \frac{2}{5}$

The curve is increasing when $\left\{x : x \in \mathbb{R}, -\frac{2}{5} < x < \frac{2}{5}\right\}$

18)

$y = 3\arctan\frac{x}{6} - 2\arctan\frac{x}{4}$

$\frac{dy}{dx} = 3 \times \frac{1}{6} \times \frac{1}{1+\left(\frac{x}{6}\right)^2} - 2 \times \frac{1}{4} \times \frac{1}{1+\left(\frac{x}{4}\right)^2}$

$= \frac{1}{2\left(1+\frac{1}{36}x^2\right)} - \frac{1}{2\left(1+\frac{1}{16}x^2\right)}$

$= \frac{1}{2+\frac{1}{18}x^2} - \frac{1}{2+\frac{1}{8}x^2}$

$= \frac{18}{36+x^2} - \frac{8}{16+x^2}$

$= \frac{18(16+x^2)-8(36+x^2)}{(36+x^2)(16+x^2)}$

$= \frac{10x^2}{x^4+52x^2+576}$

Stationary points exist when $\frac{dy}{dx} = 0$

$\Rightarrow \frac{10x^2}{x^4+52x^2+576} = 0$

$\Rightarrow 10x^2 = 0$

$\Rightarrow x = 0$

$y|_{x=0} = 3\arctan 0 - 2\arctan 0 = 0 - 0 = 0$

So there is a stationary point at $(0, 0)$

$\frac{d^2y}{dx^2} = \frac{20x(x^4+52x^2+576)-10x^2(4x^3+104x)}{(x^4+52x^2+576)^2}$

$= \frac{-20x^5+11520x}{(x^4+52x^2+576)^2}$

Points of inflection exist when $\frac{d^2y}{dx^2} = 0$

$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 0$

So the origin is a potential stationary point of inflection.

$\left.\frac{d^2y}{dx^2}\right|_{x=-0.1} = -0.00346\ldots < 0$ So the curve is concave (concave downwards) here

$\left.\frac{d^2y}{dx^2}\right|_{x=0.1} = 0.00346\ldots > 0$ So the curve is convex (concave upwards) here

As the 2nd derivative changes sign either side of the origin, we must have that the origin is a non-stationary point of inflection.

19) a)

$y = \arcsin x^2$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 2x \times \frac{1}{\sqrt{1-(x^2)^2}}$

$= \frac{2x}{\sqrt{1-x^4}}$

19) b)

$y = \arccos\sqrt{x} = \arccos\left(x^{\frac{1}{2}}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \times -\frac{1}{\sqrt{1-\left(x^{\frac{1}{2}}\right)^2}}$

$= -\frac{1}{2\sqrt{x}\sqrt{1-x}}$

$= -\frac{1}{2\sqrt{x(1-x)}}$

19) c)

$y = \arctan\frac{1}{x} = \arctan(x^{-1})$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -x^{-2} \times \frac{1}{1+\left(\frac{1}{x}\right)^2}$

$= -\frac{1}{x^2\left(1+\frac{1}{x^2}\right)}$

$= -\frac{1}{x^2+1}$

19) d)

$y = \cos(\arcsin 2x)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times \frac{1}{\sqrt{1-(2x)^2}} \times -\sin(\arcsin 2x)$

$= -\frac{2}{\sqrt{1-4x^2}} \times 2x$

$= -\frac{4x}{\sqrt{1-4x^2}}$

19) e)

$y = \sin\left(\arccos\frac{x}{2}\right)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2} \times -\frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \times \cos\left(\arccos\frac{x}{2}\right)$

$= -\frac{1}{2\sqrt{1-\frac{1}{4}x^2}} \times \frac{x}{2}$

$= -\frac{x}{4\sqrt{\frac{1}{4}(4-x^2)}}$

$= -\frac{x}{2\sqrt{4-x^2}}$

19) f)

$y = \cos(\arctan 4x)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 4 \times \frac{1}{1+(4x)^2} \times -\sin(\arctan 4x)$

Let $\theta = \arctan 4x \Rightarrow \tan\theta = 4x \Rightarrow \tan\theta = \frac{4x}{1} = \frac{\text{opp}}{\text{adj}}$

Then using a right-angled triangle, $\sin\theta = \frac{4x}{\sqrt{(4x)^2+1^2}} = \frac{4x}{\sqrt{16x^2+1}}$

$\Rightarrow \theta = \arcsin\left(\frac{4x}{\sqrt{16x^2+1}}\right)$

$\therefore \frac{\mathrm{d}y}{\mathrm{d}x} = 4 \times \frac{1}{1+(4x)^2} \times -\sin\left(\arcsin\left(\frac{4x}{\sqrt{16x^2+1}}\right)\right)$

$= -\frac{4}{16x^2+1} \times \frac{4x}{\sqrt{16x^2+1}}$

$= -\frac{16x}{(16x^2+1)^{\frac{3}{2}}}$

20)

$y = (\arcsin x)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times \frac{1}{\sqrt{1-x^2}} \times (\arcsin x)^1$

$= \frac{2 \arcsin x}{\sqrt{1-x^2}}$

$y|_{x=\frac{\sqrt{3}}{2}} = \left(\arcsin \frac{\sqrt{3}}{2}\right)^2 = \left(\frac{\pi}{3}\right)^2 = \frac{\pi^2}{9}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\frac{\sqrt{3}}{2}} = \frac{2 \arcsin \frac{\sqrt{3}}{2}}{\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}} = \frac{2 \times \frac{\pi}{3}}{\sqrt{\frac{1}{4}}} = \frac{2\pi}{3} \times 2 = \frac{4\pi}{3}$

So the equation of the tangent will be:

$y - \frac{\pi^2}{9} = \frac{4\pi}{3}\left(x - \frac{\sqrt{3}}{2}\right)$

$\Rightarrow y = \frac{4\pi}{3}x + \frac{\pi^2}{9} - \frac{2\pi\sqrt{3}}{3}$

21)

$y = \arctan(\mathrm{e}^x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^x \times \frac{1}{1+(\mathrm{e}^x)^2}$

$= \frac{\mathrm{e}^x}{1+\mathrm{e}^{2x}}$

$y|_{x=-\frac{1}{2}\ln 3} = \arctan\left(\mathrm{e}^{-\frac{1}{2}\ln 3}\right) = \arctan\left(\mathrm{e}^{\ln 3^{-\frac{1}{2}}}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=-\frac{1}{2}\ln 3} = \frac{\mathrm{e}^{-\frac{1}{2}\ln 3}}{1+\mathrm{e}^{2\left(-\frac{1}{2}\ln 3\right)}} = \frac{\mathrm{e}^{\ln 3^{-\frac{1}{2}}}}{1+\mathrm{e}^{\ln 3^{-1}}} = \frac{\frac{1}{\sqrt{3}}}{1+\frac{1}{3}} = \frac{\sqrt{3}}{4}$

So the equation of the tangent will be:

$y - \frac{\pi}{6} = \frac{\sqrt{3}}{4}\left(x + \frac{1}{2}\ln 3\right)$

$\Rightarrow y = \frac{\sqrt{3}}{4}x + \frac{\sqrt{3}}{8}\ln 3 + \frac{\pi}{6}$

22) a)

$y = x \arcsin x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = x \times \frac{1}{\sqrt{1-x^2}} + 1 \times \arcsin x$

$= \frac{x}{\sqrt{1-x^2}} + \arcsin x$

22) b)

$y = x^2 \arccos 2x$

$\frac{dy}{dx} = x^2 \times \left(2 \times -\frac{1}{\sqrt{1-(2x)^2}}\right) + 2x \times \arccos 2x$

$= -\frac{2x^2}{\sqrt{1-4x^2}} + 2x \arccos 2x$

22) c)

$y = x^3 \arctan(x^2)$

$\frac{dy}{dx} = x^3 \times \left(2x \times \frac{1}{1+(x^2)^2}\right) + 3x^2 \times \arctan(x^2)$

$= \frac{2x^4}{1+x^4} + 3x^2 \arctan(x^2)$

23)

$y = (x-5)\sqrt{-x^2+10x-24} + \sin^{-1}(x-5)$

$= (x-5)(-x^2+10x-24)^{\frac{1}{2}} + \sin^{-1}(x-5)$

$\frac{dy}{dx} = (x-5) \times \frac{1}{2} \times (-2x+10) \times (-x^2+10x-24)^{-\frac{1}{2}} + 1 \times (-x^2+10x-24)^{\frac{1}{2}} + \frac{1}{\sqrt{1-(x-5)^2}}$

$= (x-5) \times \frac{1}{2} \times -2(x+5) \times (-x^2+10x-24)^{-\frac{1}{2}} + \sqrt{-x^2+10x-24} + \frac{1}{\sqrt{-x^2+10x-24}}$

$= -\frac{(x-5)^2}{\sqrt{-x^2+10x-24}} + \sqrt{-x^2+10x-24} + \frac{1}{\sqrt{-x^2+10x-24}}$

$= \frac{1-(x-5)^2}{\sqrt{-x^2+10x-24}} + +\sqrt{-x^2+10x-24}$

$= \frac{-x^2+10x-24}{\sqrt{-x^2+10x-24}} + \sqrt{-x^2+10x-24}$

$= \sqrt{-x^2+10x-24} + \sqrt{-x^2+10x-24}$

$= 2\sqrt{-x^2+10x-24}$

So $k = 2$

24)

$y = 3x\sqrt{4-x^2} + 12\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right)$

$= 3x(4-x^2)^{\frac{1}{2}} + 12\tan^{-1}\left(\frac{x}{(4-x^2)^{\frac{1}{2}}}\right)$

$\frac{dy}{dx} = 3x \times \frac{1}{2} \times -2x \times (4-x^2)^{-\frac{1}{2}} + 3 \times (4-x^2)^{\frac{1}{2}}$

$+12 \times \frac{1\times(4-x^2)^{\frac{1}{2}} - x\times\frac{1}{2}\times -2x\times(4-x^2)^{-\frac{1}{2}}}{4-x^2} \times \frac{1}{1+\left(\frac{x}{(4-x^2)^{\frac{1}{2}}}\right)^2}$

$= -3x^2(4-x^2)^{-\frac{1}{2}} + 3(4-x^2)^{\frac{1}{2}} + \frac{12(4-x^2)^{\frac{1}{2}} + 12x^2(4-x^2)^{-\frac{1}{2}}}{4-x^2} \times \frac{1}{1+\frac{x^2}{4-x^2}}$

$= -3x^2(4-x^2)^{-\frac{1}{2}} + 3(4-x^2)^{\frac{1}{2}} + \frac{12(4-x^2)^{\frac{1}{2}} + 12x^2(4-x^2)^{-\frac{1}{2}}}{4-x^2} \times \frac{4-x^2}{4-x^2+x^2}$

$= -3x^2(4-x^2)^{-\frac{1}{2}} + 3(4-x^2)^{\frac{1}{2}} + \frac{12(4-x^2)^{\frac{1}{2}} + 12x^2(4-x^2)^{-\frac{1}{2}}}{4-x^2} \times \frac{4-x^2}{4}$

$= -3x^2(4-x^2)^{-\frac{1}{2}} + 3(4-x^2)^{\frac{1}{2}} + 3(4-x^2)^{\frac{1}{2}} + 3x^2(4-x^2)^{-\frac{1}{2}}$

$= 6(4-x^2)^{\frac{1}{2}}$

So $k = 6$

25)

$\arccos y - \arccos 3x = \frac{\pi}{3}$

$\frac{\mathrm{d}}{\mathrm{d}x}(\arccos y - \arccos 3x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\pi}{3}\right)$

$\frac{1}{\sqrt{1-y^2}}\frac{\mathrm{d}y}{\mathrm{d}x} - 3 \times \frac{1}{\sqrt{1-(3x)^2}} = 0$

$\Rightarrow \frac{1}{\sqrt{1-y^2}}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{\sqrt{1-9x^2}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3\sqrt{1-y^2}}{\sqrt{1-9x^2}}$

When $x = \frac{1}{6}$:

$\arccos y - \arccos\left(3 \times \frac{1}{6}\right) = \frac{\pi}{3}$

$\Rightarrow \arccos y - \arccos\frac{1}{2} = \frac{\pi}{3}$

$\Rightarrow \arccos y - \frac{\pi}{3} = \frac{\pi}{3}$

$\Rightarrow \arccos y = \frac{2\pi}{3}$

$\Rightarrow y = \cos\frac{2\pi}{3} = -\frac{1}{2}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\left(\frac{1}{6},-\frac{1}{2}\right)} = \frac{3\sqrt{1-\left(-\frac{1}{2}\right)^2}}{\sqrt{1-9\left(\frac{1}{6}\right)^2}} = \frac{\frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = 3$

26) a)

$y = \operatorname{cosec}^{-1} x$

$\Rightarrow \operatorname{cosec} y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosec} y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\operatorname{cosec} y \cot y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosec} y \cot y}$

$= -\frac{1}{\operatorname{cosec} y\sqrt{\operatorname{cosec}^2 y - 1}}$

$= -\frac{1}{\sqrt{x^2}\sqrt{x^2-1}}$

$= -\frac{1}{\sqrt{x^4-x^2}}$

26) b)

$y = \sec^{-1} x$

$\Rightarrow \sec y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\sec y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \sec y \tan y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec y \tan y}$

$= \frac{1}{\sec y \sqrt{\sec^2 y - 1}}$

$= \frac{1}{\sqrt{x^2}\sqrt{x^2-1}}$

$= \frac{1}{\sqrt{x^4 - x^2}}$

26) c)

$y = \cot^{-1} x$

$\Rightarrow \cot y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\cot y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\operatorname{cosec}^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosec}^2 y}$

$= -\frac{1}{1+\cot^2 y}$

$= -\frac{1}{1+x^2}$